

Gaussian vs t-Copula: Capturing Joint Crashes

Two models of “moving together.” Only one believes in contagion.

The claim we test: a correlation number tells you how two assets co-move *on average* — it says nothing about whether they crash *together*. The Gaussian copula assumes joint extremes are vanishingly rare (zero tail dependence); the t-copula assumes crashes cluster. Using two sector ETFs — Financials (XLF) and Technology (XLK) — through the GFC, COVID and 2022, we measure which model matches how markets actually behave on the worst days.

PROJECT CARD

CATEGORY

Risk Management

LIBRARIES

SciPy · NumPy · seaborn

ASSETS

XLF vs XLK (SPDR sectors)

TIMEFRAME

Jan 2007 – Dec 2024

USE

Choosing a dependence model that doesn't underestimate crash correlation (2008's lesson)

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

np.random.seed(42)
plt.rcParams["figure.dpi"] = 110
sns.set_style("whitegrid")
```

1. The disagreement

A copula separates *how assets co-move* from *how each behaves alone*. Two copulas can share the same correlation ρ yet disagree completely about the tails:

- **Gaussian copula:** as moves get more extreme, the dependence *fades*. Joint crashes are essentially coincidences. Lower-tail dependence $\lambda = 0$, always.
- **t-copula:** dependence *persists* into the extremes — crashes arrive together. $\lambda > 0$, controlled by the degrees of freedom ν (smaller ν = fatter joint tails).

This isn't academic. The Gaussian copula priced the CDOs of 2008 — and its zero-tail-dependence assumption is precisely what failed when housing defaults arrived together.

2. The contenders

	Gaussian copula	t-copula
Parameters	ρ	ρ, ν (degrees of freedom)
Lower-tail dependence λ	0 — always	$2 t_{\nu+1}(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}}) > 0$
Joint crashes	Coincidences	Expected
As $\nu \rightarrow \infty$	—	Converges to Gaussian

The t-copula *contains* the Gaussian as a limit — so if the data prefers small ν , that's the data voting for tail dependence.

3. The test

3.1 Data: XLF vs XLK through three crises

```
TICKERS = ["XLF", "XLK"]
START, END = "2007-01-01", "2024-12-31"

def load_prices(tickers, start, end):
    """Adjusted-close prices: yfinance first, Stooq as fallback."""
    try:
        import yfinance as yf
        df = yf.download(tickers, start=start, end=end, auto_adjust=True, progress=False)["Close"]
        if not df.empty:
            return df[tickers].dropna()
    except Exception as exc:
        print(f"yfinance failed ({exc}); trying Stooq...")
    cols = {}
    for t in tickers:
        url = f"https://stooq.com/q/d/l/?s={t.lower()}.us&i=d"
        cols[t] = pd.read_csv(url, parse_dates=["Date"], index_col="Date")["Close"].rename(t)
    return pd.concat(cols, axis=1).loc[start:end].dropna()

px = load_prices(TICKERS, START, END)
rets = px.pct_change().dropna()
n = len(rets)
print(f"{n} trading days, {px.index[0].date()} → {px.index[-1].date()}")
print(f"Linear correlation: {rets['XLF'].corr(rets['XLK']):.2f}")
```

3.2 Strip the marginals: pseudo-observations

Copulas work on ranks. Converting each series to its empirical percentile removes the marginal distributions and leaves only the dependence structure.

```

u = rets.rank() / (n + 1)          # pseudo-observations in (0,1)

fig, axes = plt.subplots(1, 2, figsize=(11, 4.6))
axes[0].scatter(rets["XLF"], rets["XLK"], s=4, alpha=0.4, color="steelblue")
axes[0].set_title("Raw daily returns"); axes[0].set_xlabel("XLF"); axes[0].set_ylabel("XLK")
axes[1].scatter(u["XLF"], u["XLK"], s=4, alpha=0.4, color="darkorange")
axes[1].set_title("Pseudo-observations (the copula's view)")
axes[1].set_xlabel("XLF percentile"); axes[1].set_ylabel("XLK percentile")
plt.tight_layout(); plt.show()

print("Note the crowding in the corners of the right plot – especially bottom-left (joint crashes).")

```

3.3 Fit both copulas

Gaussian: correlation of normal scores. t-copula: same ρ (via Kendall's τ), with ν chosen by maximum likelihood.

```

# --- Gaussian copula fit ---
z = stats.norm.ppf(u.values)
rho_g = np.corrcoef(z.T)[0, 1]

# --- t-copula fit: rho from Kendall's tau, nu by profile likelihood ---
tau = stats.kendalltau(u["XLF"], u["XLK"]).statistic
rho_t = np.sin(np.pi * tau / 2)

def t_copula_loglik(nu, u, rho):
    x = stats.t.ppf(u, df=nu)
    cov = np.array([[1, rho], [rho, 1]])
    num = stats.multivariate_t(loc=[0, 0], shape=cov, df=nu).logpdf(x)
    den = stats.t.logpdf(x, df=nu).sum(axis=1)
    return np.sum(num - den)

nus = np.arange(2, 31)
lls = [t_copula_loglik(v, u.values, rho_t) for v in nus]
nu_hat = nus[int(np.argmax(lls))]

fig, ax = plt.subplots(figsize=(9, 3.8))
ax.plot(nus, lls, "o-", color="steelblue")
ax.axvline(nu_hat, color="crimson", ls="--", label=f"MLE:  $\nu = \{nu\_hat\}$ ")
ax.set_xlabel("Degrees of freedom  $\nu$ "); ax.set_ylabel("Copula log-likelihood")
ax.set_title("The data votes: profile likelihood over  $\nu$ "); ax.legend()
plt.tight_layout(); plt.show()

print(f"Gaussian copula: rho = {rho_g:.3f}")
print(f"t-copula: rho = {rho_t:.3f}, nu = {nu_hat}")
print("\nSmall  $\nu$  = strong tail dependence.  $\nu$  above ~30 would mean 'basically Gaussian'.")

```

4. The scoreboard

4.1 Tail dependence: the headline number

Probability that one ETF is in its worst $q\%$ given the other is — as $q \rightarrow 0$.

```

# closed-form lower-tail dependence of the t-copula
lam_t = 2 * stats.t.cdf(-np.sqrt((nu_hat + 1) * (1 - rho_t) / (1 + rho_t)), df=nu_hat + 1)
print(f"Lower-tail dependence  $\lambda$  – Gaussian: 0.000 (by construction)")
print(f"Lower-tail dependence  $\lambda$  – t-copula: {lam_t:.3f}")

```

4.2 Empirical test: joint crash frequency

Count the days both ETFs landed in their worst 5% (and 1%) simultaneously — then ask each fitted copula how many such days it *predicts* over the same sample size.

```
def simulate_gaussian(rho, size):
    cov = np.array([[1, rho], [rho, 1]])
    z = np.random.multivariate_normal([0, 0], cov, size)
    return stats.norm.cdf(z)

def simulate_t(rho, nu, size):
    cov = np.array([[1, rho], [rho, 1]])
    g = np.random.multivariate_normal([0, 0], cov, size)
    chi = np.random.chisquare(nu, size) / nu
    x = g / np.sqrt(chi)[:, None]
    return stats.t.cdf(x, df=nu)

N_SIM = 2_000_000
u_g = simulate_gaussian(rho_g, N_SIM)
u_t = simulate_t(rho_t, nu_hat, N_SIM)

print(f"{'q':>4} {'empirical':>10} {'Gaussian':>10} {'t-copula':>10}    (joint-crash days per sample of {n})")
for q in [0.05, 0.01]:
    emp = float(((u['XLF'] < q) & (u['XLK'] < q)).mean())
    pg = float(((u_g[:,0] < q) & (u_g[:,1] < q)).mean())
    pt = float(((u_t[:,0] < q) & (u_t[:,1] < q)).mean())
    print(f"{q:>4.0%} {emp*n:>10.1f} {pg*n:>10.1f} {pt*n:>10.1f}")
```

```
# visual: simulated pseudo-observations vs reality, tails highlighted
fig, axes = plt.subplots(1, 3, figsize=(13.5, 4.4))
sets = [(u.values, "Real data (XLF/XLK)", "black"),
        (u_g[:len(u)], f"Gaussian (ρ={rho_g:.2f})", "steelblue"),
        (u_t[:len(u)], f"t-copula (ρ={rho_t:.2f}, ν={nu_hat})", "crimson")]
for ax, (dat, title, c) in zip(axes, sets):
    ax.scatter(dat[:, 0], dat[:, 1], s=3, alpha=0.35, color=c)
    ax.axvline(0.05, color="gray", ls=":", lw=0.8); ax.axhline(0.05, color="gray", ls=":", lw=0.8)
    joint = ((dat[:, 0] < 0.05) & (dat[:, 1] < 0.05)).sum()
    ax.set_title(f"{title}\njoint 5% tail: {joint} days")
    ax.set_xlabel("XLF percentile")
axes[0].set_ylabel("XLK percentile")
plt.tight_layout(); plt.show()
```

5. Verdict

The t-copula wins where it counts — the corner where portfolios die. Typical result on this pair: the Gaussian copula *underpredicts* joint 1%-tail days by a large factor, while the t-copula lands close to the observed count. The likelihood agrees: the fitted ν is small, which is the data explicitly rejecting the Gaussian limit.

Why it matters:

- **Risk models built on Gaussian dependence** understate exactly the events that cause maximum damage — simultaneous crashes across holdings. Your 99% “diversified” VaR is too optimistic in the only scenario you bought diversification for.
- **The 2008 lesson, quantified** — the Gaussian copula’s $\lambda = 0$ is the mathematical form of “national housing markets can’t all fall together.” They did. Any dependence model with zero tail dependence repeats that mistake in new clothing.

The honest caveats:

- The t-copula is *symmetric* — it implies joint booms are as likely as joint crashes. Equity data usually shows *asymmetric* dependence (crashes cluster more than rallies). A skewed-t or Clayton copula captures that; the t is the pragmatic middle ground.
- One ν for all seasons — dependence itself shifts with regimes (see our 60/40 case study). Time-varying copulas exist but multiply complexity.

Where to go next: the **Copulas & Tail Dependence** tutorial for foundations; **CVaR / Expected Shortfall** for the risk measure that consumes these joint-tail probabilities; and the CDO story in our curriculum’s credit module for the trillion-dollar version of this comparison.